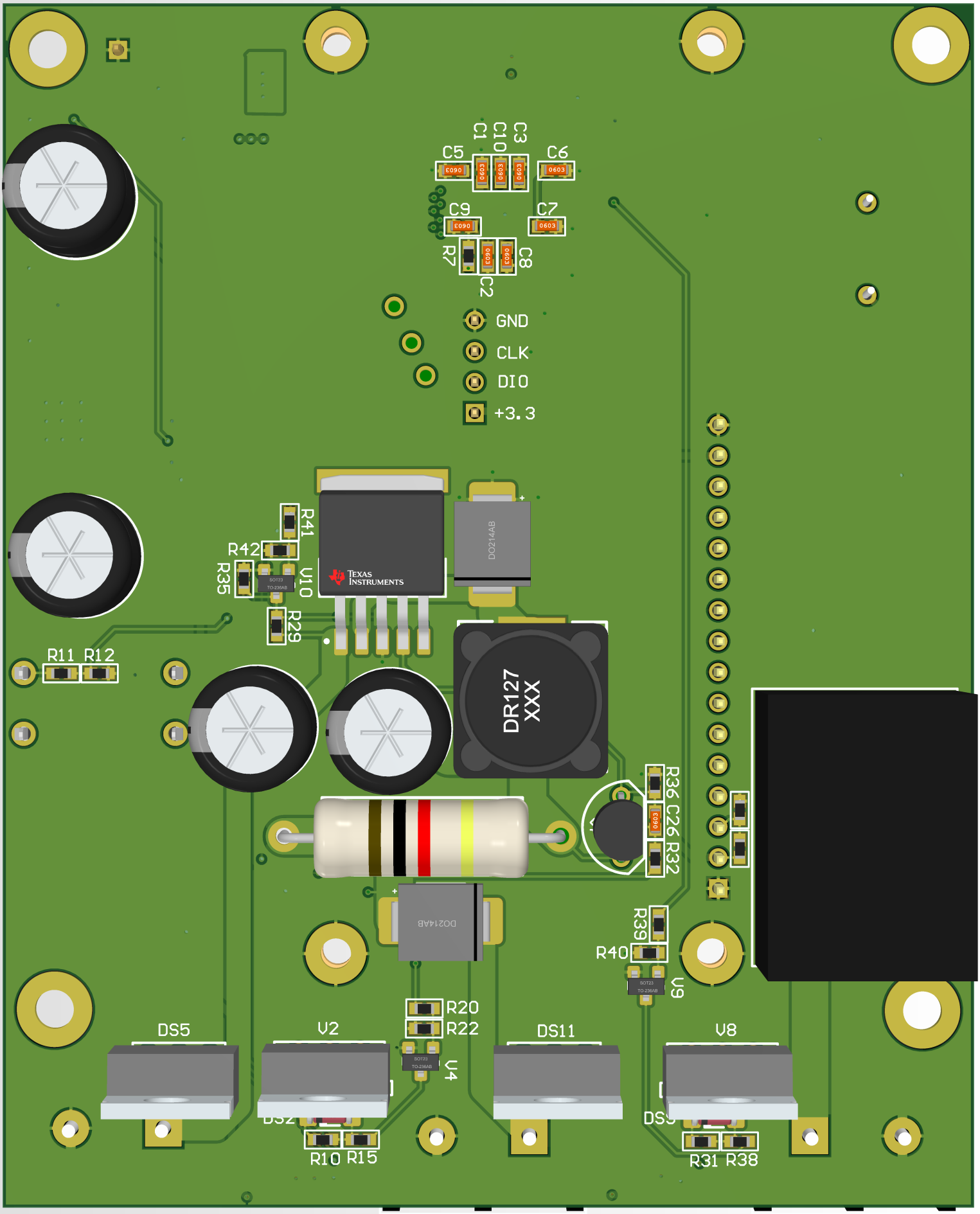
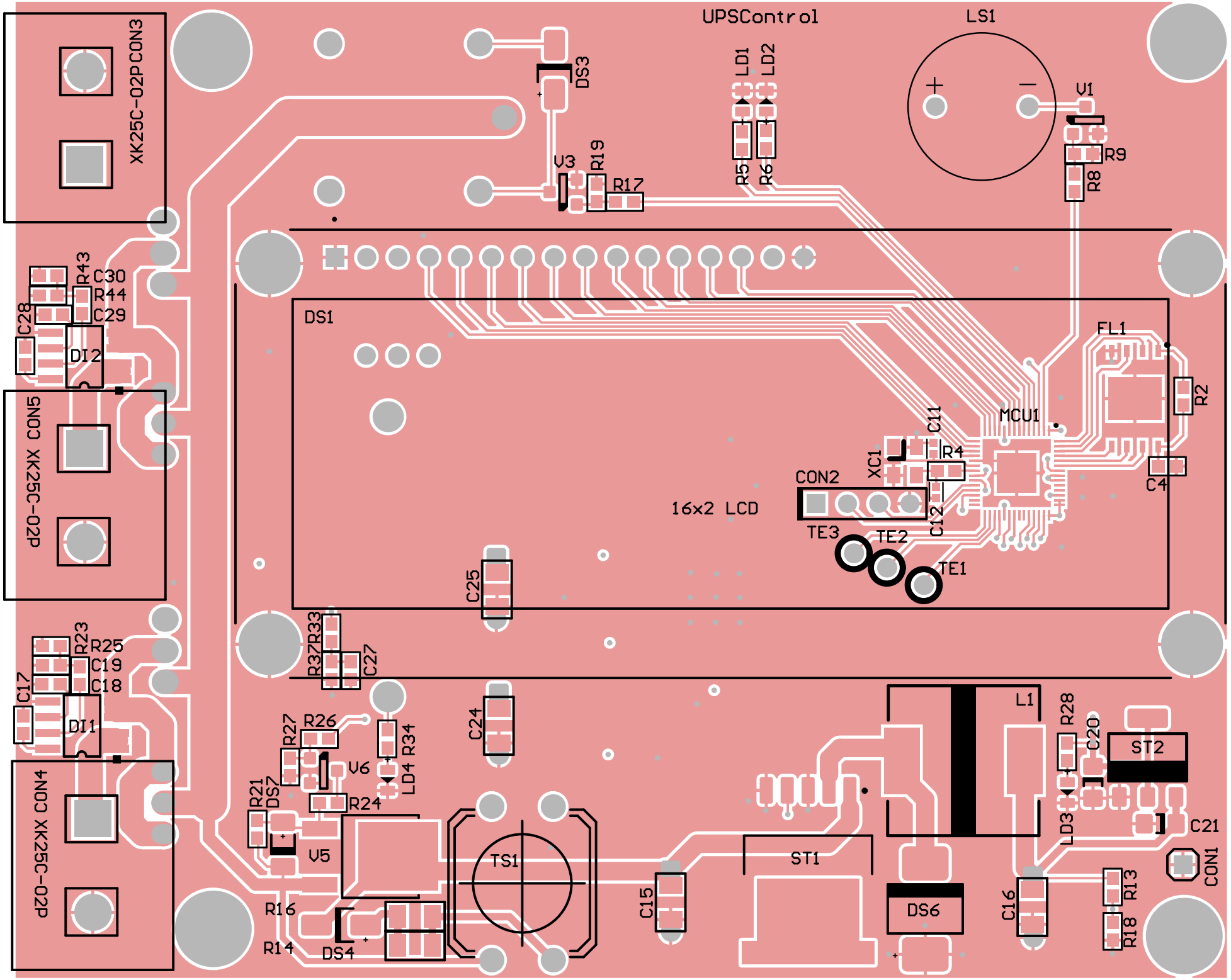
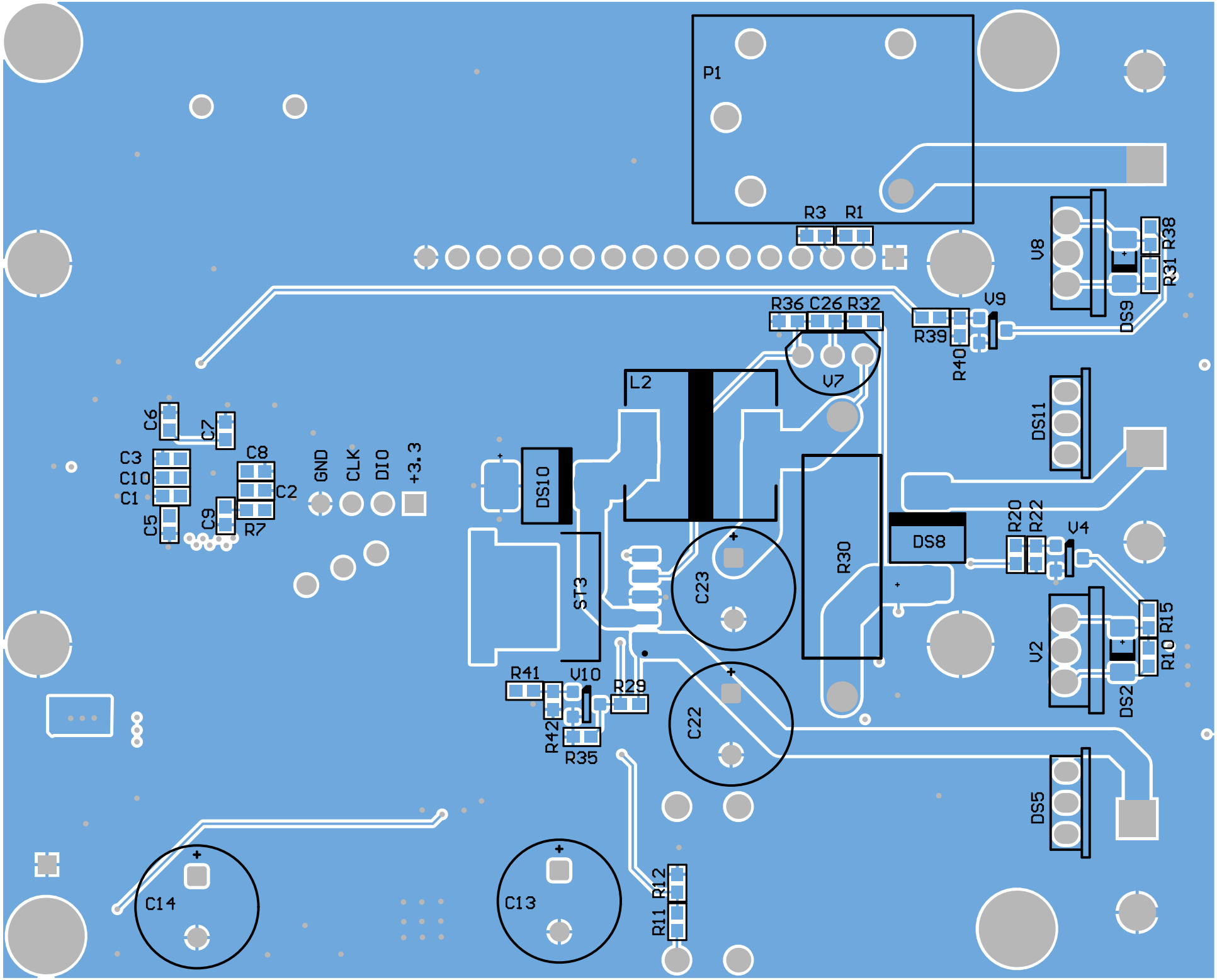
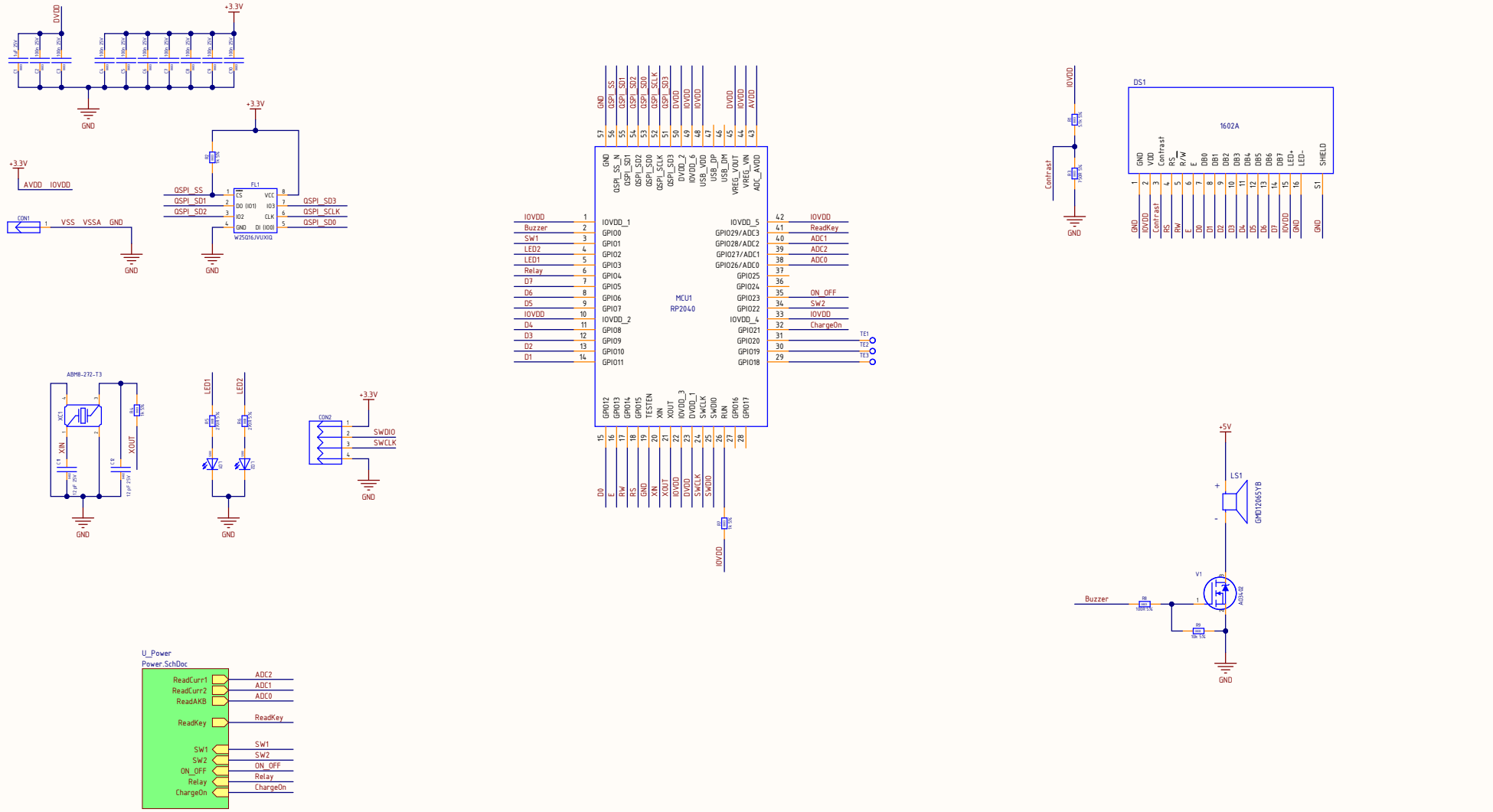
UPSControl
LS1
LD1
LD2
DS3
V1
V3
R19
R17
R5
R6
R9
R8
R43
C30
R44
C29
C28
R23
R25
C19
C18
C17
R27
R26
R34
LD4
V6
R24
R21
DS7
V5
R16
R14
DS4
C24
C15
C16
R28
C20
LD3
C21
CON1
R13
R18
DR127
XXX
TEXAS INSTRUMENTS
ON Semiconductor
FDD4141
DO214AC
DO214AB
0603
SOT23
TO-252AB
Case A

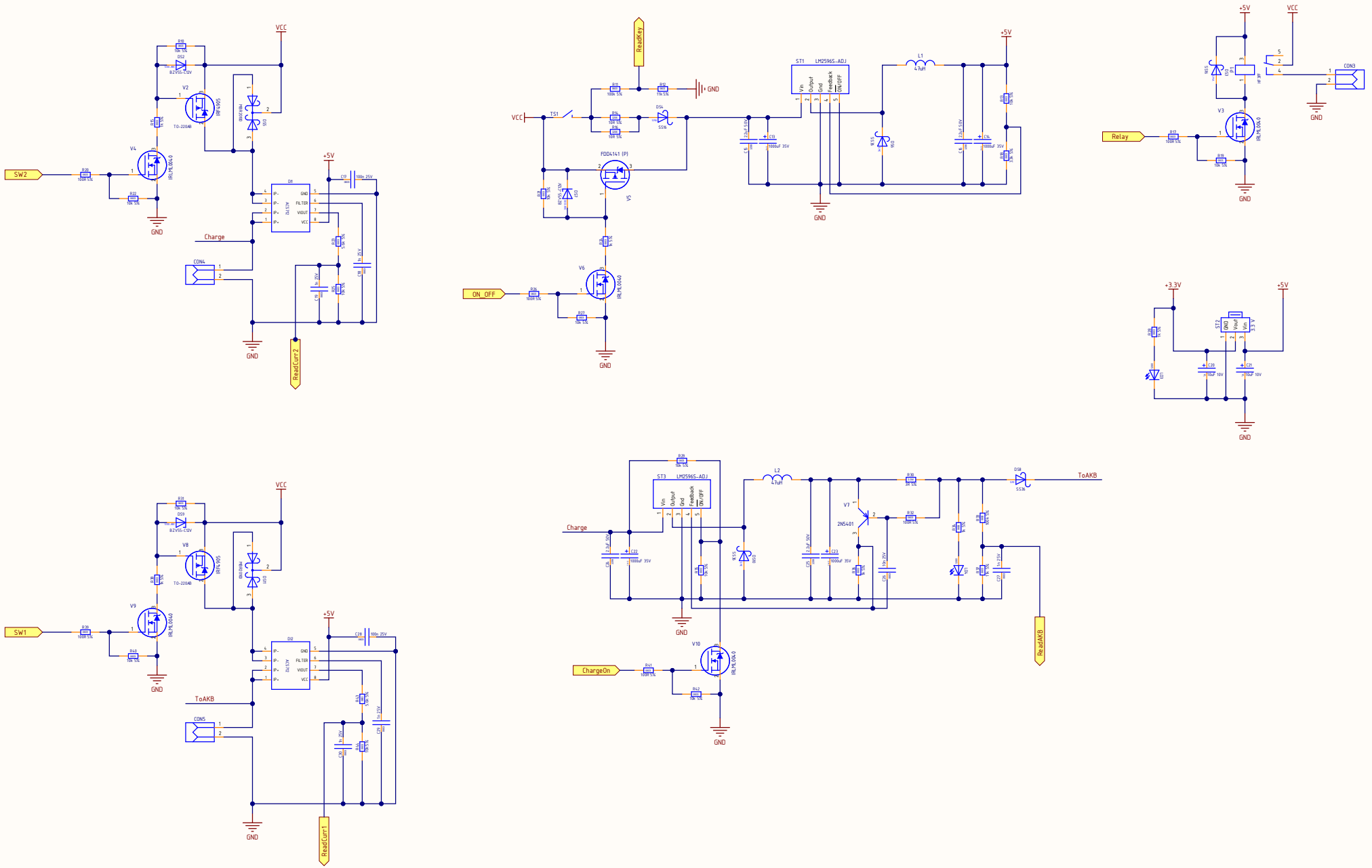

C1
C10
C3
C5
C6
C9
C7
R7
C8
C2
GND
CLK
DIO
+3.3
R41
R42
R35
V10
R29
R11
R12
DR127
XXX
R36
C26
R32
R39
R40
V9
R20
R22
V4
DS5
V2
DS11
V8
R10
R15
R31
R38


UPSControl
LS1
XK25C-02P CON3
DS3
LD1
LD2
V1
V3
R19
R17
R5
R6
R9
R8
R43
C30
R44
C29
C28
DI2
DS1
FL1
R2
MCU1
C11
XC1
R4
C4
CON2
16x2 LCD
C12
TE3
TE2
TE1
XK25C-02P CON5
C25
R23
R25
C19
C18
C17
DI1
R33
R37
C27
C24
L1
R28
C20
ST2
R27
R26
V6
LD4
R34
R21
DS7
R24
LD3
C21
CON1
V5
TS1
ST1
C15
DS6
C16
R13
R18
R16
R14
DS4
XK25C-02P CON4


P1
R3
R1
V8
R38
R31
DS9
V9
R36
C26
R32
R39
R40
V7
L2
DS11
C6
C7
C8
C3
C10
C1
C5
C9
R7
C2
GND
CLK
DIO
+3.3
DS10
ST3
C23
R30
DS8
R20
R22
V4
R41
V10
R29
R42
R35
C22
V2
R10
R15
DS2
DS5
C14
C13
R11
R12



Title	Page 1		
Size	Number		Revision
A3			V1.0
Date:	7.04.2026	Sheet of	UPSControl.PrjPcb
File	C:\Users\..\UPSControl.SchDoc	Drawn By:	R.Solovyov



Title		
Size A3	Number	Revision
Date: 7.04.2026		Sheet of
File: C:\Users\..\Power.SchDoc		Drawn By:

Designator	Footprint	Package	Quantity	Value	Link
C1	CAPC1608X09N	0603	1	1uF 25V	
C2, C3, C4, C5, C6, C7, C8, C9, C10, C17, C28	CAPC1608X09N	0603	11	100n 25V	
C11, C12	CAPC1005X06N	0402	2	12 pF 25V	
C13, C14, C22, C23	Cap-EL(10x12-5)T	10*12	4	1000uF 35V	
C15, C16, C24, C25	CAPC3216X16N	1206	4	2.2uF 50V	
C18, C19, C27, C29, C30	CAPC1608X09N	0603	5	1n 25V	
C20, C21	TANTAL_A		2	10uF 10V	
C26	CAPC1608X09N	0603	1	10n 25V	
CON1	PLS-01		1		
CON2	PLS-04		1		
CON3, CON4, CON5	XK25C-02P		3	XK25C-02P	https://voron.ua/uk/catalog/02834 3--klemmnik_barernyy_xk25c-02p_xinke_cixi_xin_electronics_factory_dg25c
DI1, DI2	SO8		2	ACS712	
DS1	LCD_RC1602B-BIW-CSV	None	1	1602A	
DS2, DS7, DS9	SOD-80C	SOD-80C	3	BZV55-C12V	
DS3, DS4	SMA	SMA	2	SS16	
DS5, DS11	TO-220A		2	MBR20100	
DS6, DS8, DS10	SMC	SMC	3	SS36	
FL1	SON127P800X600X80-9N-D		1	W25Q16JVUXIQ	
L1, L2	CDRH127NP-100MC 12*12		2	47uH	
LD1, LD2, LD3, LD4	LD0805 GREEN		4		
LS1	12mm - XDCR_AI-1223-TWT-5V-2-R	PUI Audio,	1	GMD12065YB	
MCU1	QFN40P700X700X90-57N-D		1	RP2040	https://eu.mouser.com/ProductDetail/Raspberry-Pi/SC09147?qs=T%252BzbugeAwjhSpdbCB4ve%252Bg%3D%3D
P1	JZC-20F		1	HF3FF/012-1ZS	
R1	RESC1608X08N	0603	1	5.1k 5%	
R2, R4, R7, R15, R24, R28, R34, R36, R38	RESC1608X08N	0603	9	1k 5%	
R3	RESC1608X08N	0603	1	750R 5%	
R5, R6	RESC1608X08N	0603	2	270R 5%	
R8, R17, R20, R26, R32, R39, R41	RESC1608X08N	0603	7	100R 5%	
R9, R10, R13, R19, R21, R22, R25, R27, R29, R31, R35, R40, R42, R44	RESC1608X08N	0603	14	10k 5%	
R11, R33	RESC1608X08N	0603	2	100k 5%	
R12, R37	RESC1608X08N	0603	2	11k 5%	
R14, R16	RESC3216X16N	1206	2	10R 5%	
R18	RESC1608X08N	0603	1	3.3k 5%	
R23, R43	RESC1608X08N	0603	2	5.6k 5%	
R30	RES3.00ax		1	3R 5%	
ST1, ST3	FP-KTT0005B-MFG		2	LM2596S-ADJ	
ST2	SOT-223		1	3.3 V	
TE1, TE2, TE3	КОНТ-УР		3		
TS1	SWT12-7.0		1		
V1	SOT-23		1	AO3402	
V2, V8	TO220		2		
V3, V4, V6, V9, V10	SOT-23		5	IRLML0040	
V5	DPAK		1	FDD4141 (P)	
V7	TO-92A		1	2N5401	
XC1	3.20-2.50	3.2-2.5mm	1	ABM8-272-T3	